Sentinova ThreatLens — Cyber Threat Intelligence (CTI) Platform



Sentinova ThreatLens is an enterprise-grade Cyber Threat Intelligence (CTI) aggregation, risk-scoring, and visualization platform. Designed specifically for security operations centers (SOCs) and cybersecurity leadership, Sentinova ThreatLens continuously ingests raw threat indicators from authoritative global feeds, normalizes them into atomic Indicators of Compromise (IoCs), executes an explainable multi-factor risk scoring engine, maps observables to the **MITRE ATT&CK® framework**, and serves both tactical analysts and executive stakeholders through a unified cyber-dark dual-mode interface.

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Architectural Overview

Sentinova ThreatLens bridges the gap between raw threat telemetry and operational cyber defense. Threat intelligence feeds frequently output unstructured, unranked, and duplicate data that overwhelms SOC tier-1 analysts. Sentinova ThreatLens solves this via:

1. **Autonomous Feed Ingestion**: Asynchronously harvests indicators across CISA Known Exploited Vulnerabilities (KEV), URLhaus (malware distribution and C2), and AbuseIPDB (malicious IPs and brute-force scanners).
2. **Deterministic Natural Deduplication**: Upserts records onto natural composite keys `(value, ioc_type)`, tracking recency decay and merging threat context.
3. **Explainable Composite Risk Scoring**: Calculates deterministic scores (0.0 to 100.0) with mathematical breakdowns across source reliability, threat context, and temporal recency decay.
4. **Resilient Hybrid Caching**: Integrates Redis with automatic, zero-crash in-memory TTL dictionary fallback.

5. **Dual-Mode SOC Workbench**: Instant toggle between an **Executive Strategic Landscape** (high-level risk posture, KPIs, MITRE matrices, vendor exposure) and a **SOC Analyst Tactical Workbench** (sub-second queries, bulk triage, SIEM CSV/JSON exports, and enrichment drawers).

Key Engineering Features

- **High-Concurrency SQLite WAL & Async Engine**: Configured with Write-Ahead Logging (`journal_mode=WAL`, `synchronous=NORMAL`) ensuring concurrent feed ingestion writes never lock read requests from API analysts.
- **MITRE ATT&CK® Correlation**: Automatically correlates incoming indicator tags and exploit flags to standardized technique identifiers (e.g., `T1190` Exploit Public-Facing Application, `T1105` Ingress Tool Transfer, `T1486` Data Encrypted for Impact).
- **Sub-Second Bulk Triage**: Paste up to 50 arbitrary IPs, CVEs, domains, or URLs to instantly triage known campaign assets versus clean observables.
- **SIEM Interoperability**: One-click streaming exports of filtered threat indicators in RFC 4180 compliant CSV or structured JSON for ingestion into Splunk, Microsoft Sentinel, or Elastic Security.
- **Zero-Downtime Cache Resiliency**: Employs an asynchronous facade (`app/services/cache.py`) that transparently falls back to an in-memory TTL dictionary if Redis is unreachable. Ingestion updates automatically invalidate all `threatlens:*` cache namespaces.

System Architecture & Data Pipeline

```
flowchart TD
    subgraph ExternalFeeds[External Intelligence Feeds]
        CISA[CISA KEV JSON Feed]
        URLH[URLhaus Recent API / Dump]
        ABUSE[AbuseIPDB Blacklist]
    end

    subgraph IngestionPipeline[Sentinova ThreatLens Ingestion Engine]
        SCHED[APScheduler Background Worker]
        INGEST[Async Feed Ingesters]
        RAW[(RawFeedEntry Audit Store)]
        NORM[Normalization & Upsert Pipeline]
        SCORER[Deterministic Risk Scorer & MITRE Engine]
    end

    subgraph Storage[Data & Caching Layer]
        DB[(Sentinova ThreatLens SQLite WAL / PostgreSQL)]
        REDIS[(Redis 7.0 / In-Memory Fallback)]
    end

    subgraph API[FastAPI Application Layer]
        ROUTER_F[/api/v1/feeds/]
        ROUTER_I[/api/v1/indicators/]
        ROUTER_A[/api/v1/analytics/]
    end
```

```
    subgraph UI[React 18 + Tailwind CSS Frontend]
        EXEC[Executive Strategic Dashboard]
        SOC[SOC Analyst Tactical Workbench]
        DRAWER[IOC Enrichment & Score Drawer]
        BULK[Bulk Triage Modal]
    end

    SCHED --> INGEST
    CISA & URLH & ABUSE --> INGEST
    INGEST --> RAW
    RAW --> NORM
    NORM <--> SCORER
    NORM --> DB
    INGEST -.->|Invalidate 'threatlens:*'| REDIS

    DB <--> ROUTER_F & ROUTER_I & ROUTER_A
    REDIS <--> ROUTER_I & ROUTER_A

    ROUTER_A --> EXEC
    ROUTER_I --> SOC
    ROUTER_I --> DRAWER & BULK
```

🧮 Risk Scoring & MITRE Engine

The Sentinova ThreatLens scoring algorithm calculates an audit-ready, explainable composite score:

$$\text{Composite Score} = (\text{Source Weight} \times 0.40) + (\text{Context Bonus} \times 0.35) + (\text{Recency Factor} \times 25.0)$$

1. Factor Weights

Component	Weight	Max Points	Description
Source Reliability	40%	40.0 pts	Base feed credibility (CISA KEV: 95.0, URLhaus: 80.0, AbuseIPDB: dynamic confidence scale 0–100).
Context & Threat Type	35%	35.0 pts	Threat severity multiplier: Ransomware (100.0), C2 / Command & Control (85.0), Payload Download (65.0), Exploit (55.0), Phishing (45.0), Brute Force (25.0), Scanner (15.0).
Recency Factor	25%	25.0 pts	Linear temporal decay over a 45-day window. Telemetry within 24 hours yields full 25.0 points.

2. Severity Classification Thresholds

- 🟢 **CRITICAL** (≥ 85.0): Known exploited ransomware vectors, active Cobalt Strike C2s, high-confidence malicious infrastructure.
- 🟡 **HIGH** ($65.0 - 84.9$): Standard KEV catalog exploits, verified Trojan loaders, active botnet distribution nodes.

- 🟠 **MEDIUM** ($40.0 - 64.9$): Phishing portals, credential harvesters, moderate-confidence scanning hosts.
- 🟠 **LOW** (< 40.0): Reconnaissance probes, unverified telemetry, low-confidence network noise.

💻 Technology Stack

Layer	Technologies	Key Highlights
Backend Runtime	Python 3.14 / 3.12, FastAPI, Uvicorn (ASGI)	Async I/O, PEP 649/749 typing, high throughput
ORM & Database	SQLAlchemy 2.1, SQLite 3 (WAL mode), Alembic	Mapped declarative annotations, crash-resilient batch commits
Scheduling	APScheduler 3.10	Non-blocking async interval job runner
Caching Layer	Redis 7.0 + In-Memory TTL Dict Fallback	Automatic fallback on connection drop, prefix invalidation
Frontend Framework	React 18, Vite 6	Fast HMR, optimized Rollup bundling
Styling & Icons	Tailwind CSS 3.4, Lucide React	Cyber dark SOC aesthetic (`bg-slate-950`), responsive design
Data Visualization	Recharts 2.15	Custom dark-mode tooltips, interactive donut & bar charts
Containerization	Docker, Docker Compose, Nginx Alpine	Multi-stage production build, isolated bridge network

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🗂 Repository Structure

```
Sentinova-ThreatLens/
├── docker-compose.yml              # Production container orchestration
├── README.md                       # Architecture & deployment documentation
├── backend/
│   ├── Dockerfile                  # Python 3.12-slim backend container
│   ├── requirements.txt            # Pinned production Python dependencies
│   ├── .dockerignore               # Container build exclusions
│   ├── .env.example                # Configuration template
│   ├── test_phase3.py              # End-to-end backend test & calibration suite
│   └── app/
│       ├── __init__.py
│       ├── config.py               # Pydantic v2 BaseSettings environment loader
│       ├── database.py             # SQLite WAL engine & SessionLocal factory
│       ├── main.py                 # FastAPI application, lifespan & router
mounting
│       ├── models/
│       │   ├── __init__.py
│       │   └── threat.py           # Indicator, Feed, and RawFeedEntry ORM models
```

```
│       ├── schemas/
│       │   ├── __init__.py
│       │   └── threat.py           # Pydantic validation schemas & API contracts
│       ├── services/
│       │   ├── __init__.py
│       │   ├── cache.py            # Resilient async caching with in-memory
fallback
│       │   ├── ingester.py         # Async CISA KEV, URLhaus, AbuseIPDB ingesters
│       │   ├── normalizer.py       # Deduplication, upsert & normalization
pipeline
│       │   ├── scheduler.py        # Background ingestion orchestration
│       │   └── scorer.py           # 3-factor composite risk engine & MITRE
mapper
│       └── routers/
│           ├── __init__.py
│           ├── analytics.py        # Executive KPI & threat landscape analytics
│           ├── feeds.py            # Feed telemetry, health & manual sync
triggers
│           └── indicators.py       # SOC filterable IOC table, lookup & SIEM
export
└── frontend/
    ├── Dockerfile                  # Multi-stage Node.js + Nginx Alpine build
    ├── nginx.conf                  # Static SPA server & /api/ reverse proxy
    ├── package.json                # React 18, Vite, Tailwind, Recharts
dependencies
    ├── vite.config.js              # Vite dev server with /api proxy to
localhost:8000
    ├── tailwind.config.js          # Cyber dark design system tokens
    ├── postcss.config.js
    ├── index.html                  # HTML entrypoint with JetBrains Mono & Inter
    └── src/
        ├── App.jsx                 # View state coordinator, drawer & modal
manager
        ├── index.css               # Tailwind base directives & custom cyber
scrollbars
        ├── main.jsx                # React StrictMode root entrypoint
        ├── components/
        │   ├── Navbar.jsx          # Dual-mode switcher, backend liveness & sync
button
        │   ├── IndicatorDrawer.jsx # Slide-over score breakdown & MITRE inspector
        │   ├── BulkLookupModal.jsx # 50-item batch IOC triage modal
        │   └── FeedStatusModal.jsx # Feed health registry and telemetry viewer
        ├── pages/
        │   ├── AnalystView.jsx     # SOC Tactical Workbench with pagination &
export
        │   └── ExecutiveView.jsx   # Strategic CTI posture with Recharts
visualizations
        └── services/
            └── api.js              # Centralized Axios client for all backend
routes
```

API Reference

All routes are mounted under the `/api/v1` namespace.

Method	Endpoint	Description	Key Query / Body Parameters
`GET`	`/api/v1/health`	Liveness and version verification	—
`GET`	`/api/v1/analytics/executive-summary`	300s cached executive KPIs and landscape metrics	—
`GET`	`/api/v1/indicators`	Paginated, filterable, and sortable IOC table	`page`, `page_size`, `severity`, `ioc_type`, `mitre_technique`, `search`, `sort_by`, `sort_order`
`GET`	`/api/v1/indicators/lookup`	Single IOC enrichment with transparent score breakdown	`q` (e.g. `?q=193.142.146.35` or `?q=CVE-2024-21887`)
`POST`	`/api/v1/indicators/bulk-lookup`	Batch triage up to 50 raw indicators	`{"indicators": ["1.1.1.1", "CVE-2023-1234"]}`
`GET`	`/api/v1/indicators/export`	Stream filtered indicators in CSV or JSON	`format` (`csv` or `json`), plus all table filters
`GET`	`/api/v1/feeds/`	Registered feed status, telemetry health, ingested counts	—
`POST`	`/api/v1/feeds/sync`	Trigger immediate ingestion sync and purge cache	—

Interactive Swagger documentation is available at `http://localhost:8000/api/v1/docs`.

Quick Start Guide

Option A: Docker Compose (Recommended)

Run the entire stack (FastAPI Backend, Redis Caching, Nginx React Frontend) with a single command:

```
# Clone the repository
git clone https://github.com/your-username/sentinova-threatlens.git
cd sentinova-threatlens
```

```
# Launch all containers
docker compose up --build -d
```

- **Frontend Application**: `http://localhost:5173`
- **FastAPI Documentation**: `http://localhost:8000/api/v1/docs`
- **Backend Healthcheck**: `http://localhost:8000/api/v1/health`

To stop the containers:

```
docker compose down
```

Option B: Local Native Execution

1. Backend Setup (PowerShell / Windows)

```
cd backend

# Create and activate virtual environment
python -m venv .venv
.\.venv\Scripts\Activate.ps1

# Install pinned dependencies
pip install -r requirements.txt

# Run backend test & dry-run sync
python test_phase3.py

# Launch FastAPI development server
uvicorn app.main:app --reload --host 127.0.0.1 --port 8000
```

2. Frontend Setup (Separate Terminal)

```
cd frontend

# Install Node dependencies
npm install

# Start Vite dev server with proxy to port 8000
npm run dev
```

Navigate to `http://localhost:5173` in your browser.

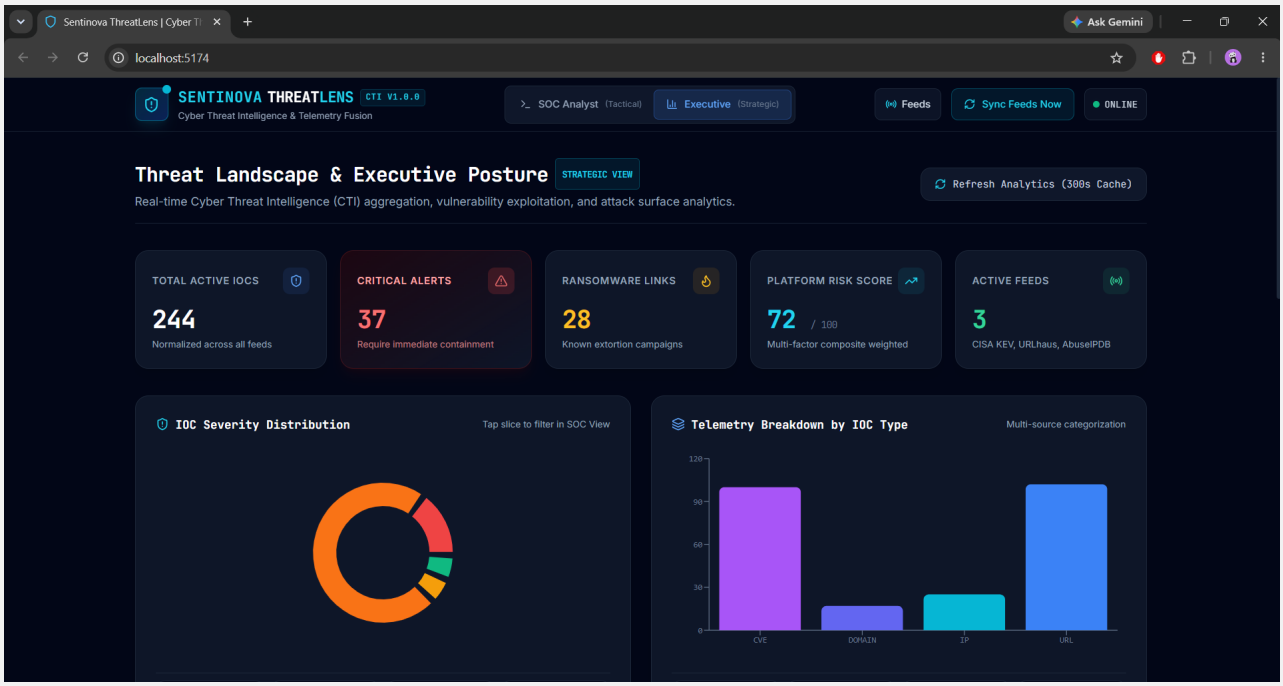
💻 Dashboard Walkthrough

1. Executive Strategic Landscape

- **Real-Time KPIs**: Total IOC inventory, active critical alerts, ransomware-linked CVEs, platform composite risk score, and healthy telemetry feeds.
- **Severity Donut**: Interactive breakdown of `CRITICAL`, `HIGH`, `MEDIUM`, and `LOW` indicators. Clicking any segment automatically transitions the analyst to the filtered tactical table.
- **MITRE ATT&CK Matrix**: Observed technique volume distribution (`T1190`, `T1105`, `T1486`, `T1071.001`, `T1110`).
- **Targeted Vendors & Origin Countries**: Surface CISA KEV vendor concentrations and geo-origin distribution.

Strategic Executive View

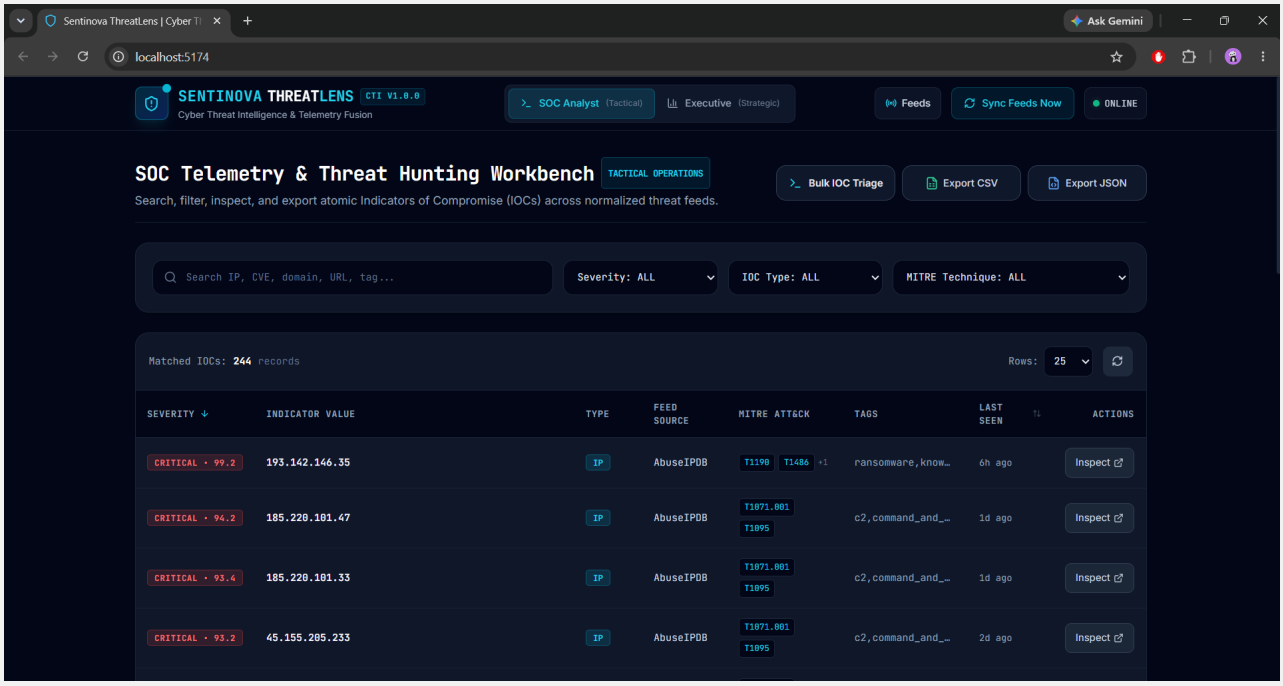
> High-level telemetry KPIs, severity distributions, top targeted vendors, and threat origin analytics.



2. SOC Analyst Tactical Workbench

- **Sub-Second Multi-Filtering**: Full-text debounced search across values, threat types, and tags; filterable by severity, IoC type, and MITRE technique.
- **Bulk IOC Triage Modal**: Ingests up to 50 observables pasted from incident tickets or SIEM alerts, partitioning them into known malicious indicators vs. clean items.
- **Slide-Over Enrichment Drawer**: Click any indicator to inspect transparent scoring progress bars, deep links to MITRE ATT&CK documentation, related attack campaign assets, and raw JSON telemetry.
- **SIEM Export**: Instant CSV and JSON downloads preserving active filter criteria for firewall or EDR blocklist deployment.

Tactical SOC Analyst Workbench

> Real-time threat hunting, multi-parameter filtering, atomic IOC lookup, and SIEM exports.
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> 
>

Author & Security Profile

Sentinova ThreatLens was engineered by **Abhijeet Sharma** as a demonstration of production-grade Cyber Threat Intelligence engineering, deterministic risk analysis, and full-stack cybersecurity application architecture.

- **Focus Areas**: Cyber Threat Intelligence (CTI), Detection Engineering, Threat Hunting, Full-Stack Python/React Systems.
- **Connect**: LinkedIn • GitHub • Security Portfolio

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